

Q.

B	A	d	d-d	(d-d) ²
110	120	-10	-6.84	70.56
20	118	2	3.6	12.96
125	125	0	1.6	2.56
132	136	-4	-2.4	5.76
125	121	4	5.6	31.36

$$S = \frac{\sum (d-d)^2}{n-1}$$

$$S = \frac{123.2}{4} = 5.54$$

$$\bar{d} = -1.6$$

Test whether significance at 1% LOS.

$$|t| = \frac{-1.6}{5.54/\sqrt{5}} = 0.64$$

$$\left\{ |t| = \frac{\bar{d}}{S/\sqrt{n}} \right.$$

Q. Null Hypothesis says that

9/3/26 Module 4: Anomaly detection.

Outliers

1. Uni-variate
 2. Multi-variate
- Analysis.

Type: global, contextual, collective.

Outlier Treatments

1. Trimming
2. Discretization
3. capping (winsorization)
4. imputation (cant write)

10/3/26

Outlier detection:

10/3/26 Q. IQR

24, 27, 24, 28, 35, 26, 31, 53, 41, 64, 29

1. Sort in Ascending order

2. Q1, Q3

3. IQR = Q3 - Q1

4. UB = Q3 + 1.5 IQR

5. LB = Q1 - 1.5 IQR

6. Outliers

steps.

Solution:

22, 24, 26, 28, 29, 31, 35, 37, 41, 53, 64

1. Q1 = $\left(\frac{n+1}{4}\right)^{th} = 3^{rd} = 26$

2. Q3 = $3\left(\frac{n+1}{4}\right)^{th} = 9^{th} = 41$

3. IQR = 41 - 26 = 15

4. UB = 41 + 1.5(15) = 63.5

5. LB = 26 - 1.5(15) = 3.5

6. outliers = 64

Problems:

1. Does not suit well with multivariate
2. Restricted bound of 1.5 no flexibility.

Q. Z-score (mainly with normally distributed data)

1. Calculate μ

2. calculate SD

3. Zscore $Z = \frac{x - \mu}{\sigma}$

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{n}}$$

4. Set the threshold $|Z| > 3$ $|Z| > 2.5$

$|Z| > 2$

can assume all/
write conclusion

Q. 10, 22, 24, 25, 26, 28, 26

Solution:

$$\mu = \frac{\sum x}{n} = \frac{200}{7} = 28.57$$

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{n}}$$

Let Assumed mean be 28

x	d	d ²	z (shown calculation)
10	-18	324	-1.13
22	-6	36	-0.37
24	-4	16	-0.252
25	-3	9	-0.189
26	-2	4	-0.1263
28	0	0	0
65	37	1369	2.33
	4	1758	

$$\sigma = \sqrt{\frac{\sum d^2}{N} - \left(\frac{\sum d}{N}\right)^2}$$

$$\sigma = \sqrt{\frac{1758}{7} - \left(\frac{4}{7}\right)^2}$$

$$\sigma = 15.83$$

z score = $\frac{x - \mu}{\sigma}$ (for all points)

μ (use 28.57)

8. 160, 162, 163, 165, 167, 170, 173, 178, 180, 300
Threshold = 3.

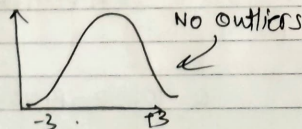
Solution:

$$\mu = 1818/10 = 181.8 \text{ (Assumed be 182)}$$

x	d	d ²	z score
160	-22	484	-0.54
162	-20	400	-0.496
163	-19	361	-0.47
165	-17	289	-0.42
167	-15	225	-0.37
170	-12	144	-0.29
173	-9	81	-0.22
178	-4	16	-0.095
180	-2	4	-0.045
300	118	13924	2.96

$$\sigma = \sqrt{\frac{15928}{10} - \left(\frac{-2}{10}\right)^2}$$

$$\sigma = 39.90$$



③ Outlier detection using Distance based method.

Algorithm:

Input: Dataset D, parameter K

Algo: 1. Each point P_i in the dataset D

- (i) calculate the distance of P_i to all other points
- (ii) sort the distance in ascending order
- (iii) select K smallest distance
- (iv) calculate the outlier score.

2. Rank all the points in score by descending order.

3. point with highest score is outlier

	Q: (2,3)	(3,4)	(3,3)	(4,4)	(3,2)	(15,15)
P1 (2,3)	0	1.41	1	2.23	1.41	17.69
P2 (3,4)	1.41	0	1	1	2	16.27
P3 (3,3)	1	1	0	1.41	1	16.97
P4 (4,4)	2.23	1	1.41	0	2.23	15.55
P5 (3,2)	1.41	2	1	2.23	0	17.69
P6 (15,15)	17.69	16.27	16.97	15.55	17.69	0

K=3 ∴ 3 Nearest Neighbours

Pi	P3, P2, P5	1, 1.41, 1.41	1.27 (4)
P2	P3, P4, P1	1, 1, 1.41	1.13 (5)
P3	P1, P2, P5	1, 1, 1	1 (6)
P4	P2, P1, P5	1, 2.23, 2.23	1.82 (2)
P5	P3, P1, P2	1, 1.41, 2	1.47 (3)
P6	P4, P2, P3	15.55, 16.27, 16.97	16.26 (1)
points	KNN	KNN	Outlier
	Nearest	Distance	score

exclude P1-P1 nearest point distance

highest rank - 16.26
Ans: $P_6(15, 15)$ is an outlier.

Rank Matrix

1	P_6	16.26	outlier
2	P_4	1.54	NO
3	P_5	1.47	NO
4	P_1	1.27	NO
5	P_2	1.3	NO (centre)
6	P_3	1	NO (centre)

NO outlier one always be

g. $(1, 2)$ $(2, 2)$ $(2, 3)$ $(3, 2)$ $(8, 8)$ $k=2$

P_1 $(1, 2)$	$(1, 2)$	$(2, 2)$	$(2, 3)$	$(3, 2)$	$(8, 8)$
P_2 $(2, 2)$	0	1	1.41	2	9.21
P_3 $(2, 3)$	1	0	1	1	8.48
P_4 $(3, 2)$	1.41	1	0	1.41	7.81
P_5 $(8, 8)$	2	1	1.41	0	7.81
	9.21	8.48	7.81	7.81	0

($k=2$)

Points	KNN node	KNN Distance	Outlier score
P_1	P_2, P_3	1, 1.41	1.20 (4)
P_2	P_1, P_3	1, 1	1 (5)
P_3	P_2, P_1	1, 1.41	1.20 (3)
P_4	P_2, P_3	1, 1.41	1.20 (2)
P_5	P_4, P_3	7.81, 7.81	7.81 (1)

Rank Matrix:

1	P_5	7.81	Outlier
2	P_4	1.20	NO
3	P_3	1.20	NO
4	P_1	1.20	NO
5	P_2	1	NO (centre)

Ans: $P_5(8, 8)$ is the outlier.
 centre P_2

g. $(1, 1)$ $(1, 0)$ $(3, 2)$ $(2, 0)$ $k=2$

	$(1, 1)$	$(1, 0)$	$(3, 2)$	$(2, 0)$
P_1	$(1, 1)$	0	1	2.23
P_2	$(1, 0)$	1	0	2.82
P_3	$(3, 2)$	2.23	2.82	0
P_4	$(2, 0)$	1.41	1	2.23

Points	KNN node	KNN distance	Outlier score
P_1	P_2, P_4	1, 1.41	1.20 (2)
P_2	P_1, P_4	1, 1	1 (4)
P_3	P_1, P_4	2.23, 2.23	2.23 (1)
P_4	P_2, P_1	1, 1.41	1.20 (3)

Rank Matrix

1	P_3	2.23	Outlier
2	P_1	1.20	NO
3	P_4	1.20	NO
4	P_2	1	NO (centre)

Ans:
 P_3 is an outlier;
 P_2 is centre.

g. sepal length

sepal length	sepal width	species	Distance
5.3	3.7	setosa	0.60
5.1	3.8	S	0.60
7.2	3.0	V	0.70
5.4	3.4	S	2.03
5.1	3.3	S	0.36
5.4	3.9	S	0.22
7.4	2.8	ver.	0.82
6.1	2.8	veris	2.22
7.3	2.9	V	0.94
6.0	2.7	V C	2.10
5.8	2.8	V	0.89
6.3	2.3	V C	0.67
5.1	2.5	V C	1.36

6.3	2.5	VC	1.25
5.5	2.4	VC	0.76

SL SW spec?
5.2 3.1 s.

5NN = 0.22, 0.36, 0.60, 0.60, 0.70
s s s VC s.

Q2 - Maths	CS	Rank	Distance	3NN =
4	3	F	5.38	
6	7	P	1	1, 1, 2
7	8	P	1.00	P, P, P.
5	5	F	3.16	
8	8	P	2	Paul

math 6 cs 8 = (6, 8)

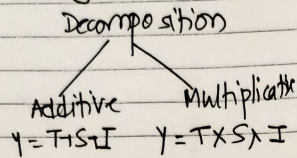
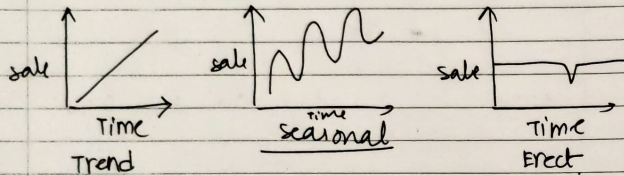
//
16/3/26

Imp: Taxonomy of time series forecasting

30/3/26 Modules: Time series Forecasting

Components of time series

- 1) Trend / Secular trend
- 2) seasonality
- 3) cyclic
- 4) Irregularity



Decomposition based forecasting

$T = 33 - 29.5 = 0.5$

Q. Year	Quarter	Sale (Y)	4MA	2MA (T)	S = Y - T
1	Q1	18			
	Q2	20			
	Q3	34	29.5	29.5	4.5
	Q4	44	30	29.5	14
2	Q1	22	30	30	-8.5
	Q2	20	31	30.5	-11.5
	Q3	38	32	31.5	6.5
	Q4	48	31	31.5	17
3	Q1	18	31	31.5	-13.5
	Q2	26	32	31.5	-13
	Q3	42	34	33.7	
	Q4	56			
4	Q1				
	Q2				
	Q3				
	Q4				

$$Y = T + I$$

$$T = A - MO$$

$$Y = T + S$$

$$S = Y - T$$

Seasonal

Year	Q1	Q2	Q3	Q4
1			1.5	14
2	-8.5	-11.5	6.5	17
3	-13.5	-13		
	-11	-12.25	5.5	15.5

$$\text{Year 4 Q4} = 33 + (6 \times 0.5) = 36 (T)$$

$$S = 15.5$$

$$\text{forecast} = 51.5$$

$$\text{Year 4 Q2} = 33 + 4(0.5) = 35$$

$$F = 35 - 12.25 = 22.75$$

this entire part must be total

Q. Year	Quarter	(Demand) Sale	Quantity sum	2 Period	MA (T)	S (Y - T)
1987	Q1	74				
	Q2	76				
	Q3	74	304.7	208/8	38.5	35.5
	Q4	80	312	208/8	38.5	41.5
1988	Q1	82	304.7	292/8	36.5	45.5
	Q2	68	280	271/8	33.87	34.1
	Q3	50	262	256/8	32	18
	Q4	62	250	253/8	31.6	30.4
1989	Q1	70	256	266/8	33	36.7
	Q2	74	276	284/8	35.5	38.4
	Q3	70	296			
	Q4	82				
					$T = 35.8 - 38.5 = -0.38$	

- ① Decomposition
- ② Smoothing
- ③ Regression

seasonal matrix

Year	Q1	Q2	Q3	Q4
1			35.5	41.5
2	45.5	34.1	18	30.4
3	36.7	38.2		
	41.1	36.15	26.75	35.95

will change

$$\text{Year } 1990 - 94 \quad f = 71.5 + 6x(-0.78) + 0.85$$

$$= 67.695$$

$$\text{Year } 1991 - 92 \quad f = 71.5 + 7x(-0.78) + 1.375$$

$$= 67.415$$

Q.	2020	2021	2022	
Q1	15	20	15	Additive
Q2	22	18	22	
Q3	32	26	40	
Q4	40	46	54	

Year	Quarter	Value	4MA	2MA	S = Y - T
2020	Q1	15			
	Q2	22	27.25		
	Q3	32	28.5	27.9	4.1
	Q4	40	27.5	28	12
2021	Q1	20	28.5	28	-8
	Q2	18	30	29.3	-11.3
	Q3	36	28.8	29.4	6.6
	Q4	46	29.8	29.3	16.7
2022	Q1	15	30.8	30.3	-15.3
	Q2	22	32.8	31.8	-9.8
	Q3	46			
	Q4	54			

AT-2 7.5

seasonal matrix

Year	Q1	Q2	Q3	Q4
2020			4.1	12
2021	-8	-11.3	6.6	16.7
2022	-15.3	-9.8		
	-11.65	-10.55	5.35	14.35

$$T = \frac{31.8 - 27.9}{7} = 0.55$$

$$2025 Q1 = 31.8 + 14(0.55) - 11.65 = 26.2$$

$$\left(\begin{array}{l} 2022 - Q3, Q4 \\ 2023 - 4 \\ 2024 - 4 \\ 2025 - 1 \end{array} \right)$$

in multiplication only : $Y = TXC$ $S = Y/T$
for forecast multiply

6/4/26 Smoothing (forecasting)

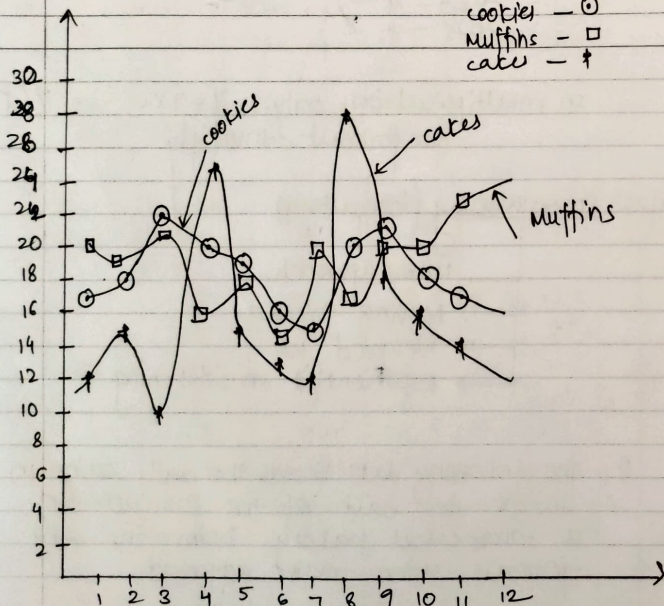
- Naive approach
- Moving Average
- Weighted MA
- Exponential smoothing

Q. The following data shows the daily sales in dozens for each of the product in a commercial bakery. Determine the forecast using naive approach.

Drawback of smoothening: Does not address
 naive approach.
 seasonal.

Day	cookies	Muffins	cakes
1	17	20	12
2	18	19	15
3	22	21	10
4	20	16	(25)
5	19	18	15
6	16	15	13
7	15	20	12
8	20	19	(28)
9	21	20	18
10	18	22	16
11	17	23	14
12	? (17)	? (23)	? (28)

check



latest method 1 cant
 0.5 0.3 0.2
 Weight = 1 & 0.4 by
 (F) (F) (F) (F) divide weight (F)
 2MA (last 2 years)
 3MA
 3WMA

8. Year	Actual Demand	Naive	(F)	(F)	(F)	(F) divide weight (F)	Exp
1	310	310	—	—	—	—	310
2	365	365	—	—	—	—	365
3	395	395	337.5	—	—	—	337.5
4	415	415	380	365.6	389.1	389.1	380.32
5	450	450	405	391.6	399	399	380.32
6	465	465	432.5	420	428.5	428.5	408.192
7	?	465	457.5	443.3	450.5	450.5	430.91

step 1

Formula:

- Naive: $F_{t+1} = A_t$
 - $MA = F_{t+1} = (A_t + A_{t-1} + \dots + A_{t-n}) / n$
 - $WMA = F_{t+1} = (aA_t + bA_{t-1} + cA_{t-2} + \dots) / (a+b+c)$ (weights)
 - $ES = F_{t+1} = \alpha A_t + (1-\alpha) F_t$
- ↑ previous Actual year ↑ Forecast previous

step 2

Error	Naive	2MA	3MA	3WMA	Exp
1	—	—	—	—	0
2	55	—	—	—	55
3	20	57.5	—	—	63
4	20	35	58.4	46	57.8
5	35	45	58.4	51	69.68
6	15	32.5	45	36.5	56.808
7	—	—	—	—	—
MAE:	131.5 = 42.3	42.5	53.9	44.5	50.38

$$MAE = \frac{\sum |A_t - F_t|}{n}$$

$$MSE = \frac{\sum (A_t - F_t)^2}{n}$$

$$RMSE = \sqrt{\frac{\sum (A_t - F_t)^2}{n}}$$

$$MAPE = \frac{\sum \frac{|A_t - F_t|}{A_t} \times 100}{n}$$

9. Only Exponential: $\alpha = 0.9$

higher alpha
smoother value
clear error.

Year	Demand	Exp(f)	Error
1	13	13	0
2	17	13	4
3	19	16.6	2.4
4	23	18.76	4.24
5	24	22.576	1.424
6	2	23.857	—
			(15)
			MAE =
			2.910

Regression

y.

forecast

time	Sales	xy	x ²	trend	prior
1	600	600	1	801.4	201.4
2	1550	3100	4	1161	389
3	1500	4500	9	1520.6	203.9
4	1500	6000	16	1880.2	380.2
5	2400	12000	25	2239.8	160.2
6	3100	18600	36	2599.4	500.6
7	2600	18200	49	2459	359
8	2900	23200	64	3318.6	418.6
9	3800	34200	81	3678.2	121.8

10	4500	45000	100	4037.8	462.2
11	4000	44000	121	4397.4	397.4
12	4900	58800	144	4757	143
Σ	278	33350	650	4757	3553.4/12

$$y = a + bx$$

$$a = \frac{\sum y}{n} - \frac{b \sum x}{n}$$

$$b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$= 296.14$$

(slight difference)

$$= 296.17$$

$$b = \frac{12 \times 268200 - 78 \times 33350}{12 \times 650 - (78)^2} = 359.61$$

$$a = \frac{22350}{12} - \frac{359.61 \times 78}{12} = 441.76$$

$$y = 441.76 + 359.61x$$

$$MAE = 296.17$$

calculator MSE, RMSE, MAPE also

Forecast:

when $x = 21$

$$y = 441.8 + 359.6(21)$$

$$y = 7993.4$$

error will be ± 296.17

around assumption

92. Seasonal Trend 9.

Seasonality Based Regression → in exam.

$\frac{\text{sum}}{n} \times 100$

	X	sale	xy	x ²	Trend	S
Y1	1	120	120	1	136.6	87.84
	2	150	300	4	147.5	101.69
	3	180	540	9	158.4	113.63
	4	210	840	16	169.3	124.04
Y2	5	140	700	25	180.2	77.69
	6	170	1020	36	191.1	88.95
	7	200	1400	49	202.2	99.00
	8	230	1840	64	212.9	108.02
	Σ	36	1400	6760	204	

forecast = $\frac{\text{Trend} \times \text{seasonality}}{100}$

Q predict for Q4 of Y3

$$b = \frac{8 \times 6760 - 36 \times 1400}{8 \times 204 - (36)^2} = 10.9$$

$$a = \frac{\Sigma y}{n} - \frac{b \Sigma x}{n} = \frac{1400}{8} - 10.9 \times \frac{36}{8}$$

$$a = 125.7$$

Regression Equation:

$$Y = 125.7 + 10.9X$$

(Y18D) (Y28D)

$$Q1 = \frac{87.8 + 77.69}{2} = 82.745$$

$$Q2 = \frac{101.69 + 88.95}{2} = 95.32$$

$$Q3 = \frac{113.63 + 99.00}{2} = 106.315$$

summary of chapter.

Time series

Decomposition

→ Analysis (breaking down) noise etc (modelling)

→ prediction.

(no. curvilinear)

Smoothing

→ prediction only

(no analysis)

- ① Data must be stationary
- ② Data must be linear
- ③ Data must be inverted

Regression

→ seasonality
→ Analysis (modelling)
prediction

→ ARIMA (no numerical)

sum continued →

seasonality index $\left\{ \begin{array}{l} Q4 = \frac{124.04 + 108.03}{2} = 116.035 \end{array} \right.$

for Q4 Y3 $X = 12$

$$T = 125.7 + 10.9(12) = 256.5$$

$$\text{forecast} = \frac{256.5 \times 116.035}{100}$$

$$(Y3-Q4) = 297.629$$

ARIMA

Auto Regression Integrated Moving Average

partial auto correlation Auto correlation

PACF
→ $P=0$

ACF

1932

$$AR(p) \rightarrow Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \phi_3 Y_{t-3} + \dots + \phi_p Y_{t-p} + \epsilon_t$$

0.2 m $\rightarrow I(d) \rightarrow Y_t^1 = Y_t - Y_{t-1}$

1940

$$MA(q) \rightarrow Y_t = c + \epsilon_t + \psi_1 \epsilon_{t-1} + \psi_2 \epsilon_{t-2} + \dots + \psi_q \epsilon_{t-q}$$

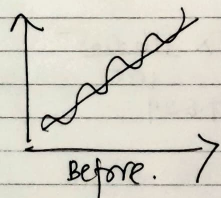
ARMA(p, q)

(if data is not stationary) this method of ARMA failed

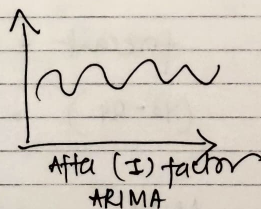
solution!

ARIMA(p, d, q)

stationary data: Mean & variance of that data must be constant



Before.



After (I) factor
ARIMA

Eq:

8.

Time	Sale
Jan	200
Feb	210
March	225
April	240
May	260
June	?

$$c = 2$$

$$\phi = 0.5$$

$$\psi = 0.3$$

ARIMA(1, 1, 1) Model.

$$ARIMA(1, 1, 1) = c + \phi Y_{t-1} + \psi \epsilon_{t-1}$$

Time	Sale	1 st diff ($Y' = Y_t - Y_{t-1}$)	
Jan	200		
Feb	210	210 - 200 = 10	
March	225	225 - 210 = 15	10 - 15 = 5
April	240	240 - 225 = 15	15 - 15 = 0
May	260	260 - 240 = 20	20 - 15 = 5
June	?		Don't consider 0 might create problem.

stationary difference is ok 5.

$$\text{maximum difference} = 5 \quad 2 + (5)$$

$$Y_{t-1} = 20$$

$$= 2 + 0.5(20) + 0.3 \times 5$$

$$= 13.5$$

appropriate model

Steps involved in ARIMA

- 1) Identification
- 2) Estimation
- 3) Diagnosis
- 4) Forecast

AIC, BIC, Adj R², sig model

$d = 0, 1, 2$

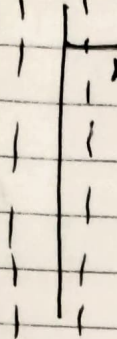
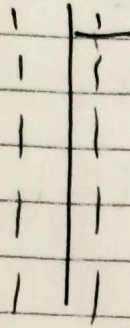
P, q can be zero

1/1

plot in "correlogram" (But first make data stationary)

(1) ACF

PACF(p)



(If point crossing line then it has significant lag)

ACF significant

ARIMA

Moving Average